



BIODIVERSITY



What is Biodiversity?

- Biodiversity describes the richness and variety of life on earth. It is the most complex and important feature of our planet. Without biodiversity, life would not sustain.
- The term biodiversity was coined in 1985. It is important in natural as well as artificial ecosystems. It deals with nature's variety, the biosphere. It refers to numbers, variety and variability of living organisms and ecosystems.
- It also covers diversity within species, between species, as well as the variations among ecosystems.
- Biodiversity holds ecological and economic significance. It provides us with nourishment, housing, fuel, clothing and several other resources. It also extracts monetary benefits through tourism. Therefore, it is very important to have a good knowledge of biodiversity for a sustainable livelihood.
- Biodiversity deals with the degree of nature's variety in the biosphere. This variety can be observed at three levels-
 - genetic diversity
 - Species diversity
 - Ecosystem diversity

Types of Biodiversity

- There are the following three different types of biodiversity:
- Genetic Biodiversity
- Species Biodiversity
- Ecological Biodiversity

Species diversity

Species diversity refers to the variety of different types of species found in a particular area. It is the biodiversity at the most basic level. It includes all the species ranging from plants to different microorganisms.

No two individuals of the same species are exactly similar. For example, humans show a lot of diversity among themselves.

Genetic diversity

It refers to the variations among the genetic resources of the organisms. Every individual of a particular species differs from each other in their genetic constitution. That is why every human looks different from each other. Similarly, there are different varieties in the same species of rice, wheat, maize, barley, etc.

Ecological diversity

An ecosystem is a collection of living and non-living organisms and their interaction with each other. Ecological biodiversity refers to the variations in the plant and animal species living together and connected by food chains and food webs.

It is the diversity observed among the different [ecosystems](#) in a region. Diversity in different ecosystems like deserts, rainforests, mangroves, etc., include ecological diversity.

- SPECIES DIVERSITY-refers to the number of plant and animal present in a community or an ecosystem
- It varies a great deal between ecosystem
- Species diversity is very high in tropical rain forests and coral reefs and low in isolated islands
- ECOSYSTEM DIVERSITY-is the variety of habitats found in an area.It refers to the variety of forest,grasslands, aquatic ecosystemsetc.

- GENETIC DIVERSITY-variety in the genetic makeup of individuals within a species
- Genetic variability is essential for a healthy breeding population of a species
- Each member of any animal or plant species differs widely from other individuals in its genetic make up,owing to the large number of combinations possible in the genes that give every individual specific characteristics

BIODIVERSITY HOT SPOTS

A Biodiversity hotspot is a biogeographic region with significant levels of biodiversity that is threatened by human activities.

Regions where exceptional concentrations of endemic species are undergoing an exceptional loss of habitat.

Concept developed by Norman Myres in 1988

Four of the world's 36 biodiversity hotspots are located in India: **The Himalayas, Western Ghats, Indo-Burma area, and Sundaland**. Two of these, the Indo-Burma area and Sundaland, are distributed throughout South Asia and are not precisely contained within India's formal borders.

BIOGEOGRAPHICAL CLASSIFICATION OF INDIA

- India has mega diversity in terms of biogeography. India has only 2.4 % of the area of land in the world but has 8% of the biological diversity of the world's content. India is divided mainly into 10 biogeographical zones. These are -
 - Trans-Himalayan Region
 - Himalayan Zone
 - Indian Desert Zone
 - Semi-Arid Region
 - Western Ghats
 - Deccan Plateau
 - Gangetic Plain
 - North East Region
 - Coastal Region
 - Andaman and Nicobar Islands.

- **What is Endemic Species?**
- Endemic species are those that are found in just one region and nowhere else in the world. For example, kangaroos are originally endemic to Australia and are found nowhere else in the world. The cases where they have been spotted outside their natural habitat is due to humans introducing them when the animal was in captivity.
- There are also other marsupials that are endemic only to Australia and its surrounding islands. The Tasmanian Tiger is one such animal that was endemic to Australia, Tasmania and New Guinea. But now, it is extinct.

- **Endemic Species of India**
- A list of the endemic species of India is mentioned below:
- **Asiatic Lion, Gir Forest**
- Asiatic Lion is also known as the Indian Lion and can be only found in and around Gir Forest National Park of Gujarat. These are listed as endangered species. These are one of the five big cats found in India, the others being Indian Leopards and Bengal Tigers.
- **Kashmir Stag, Kashmir Valley**
- Also known as Hangul, the Kashmir Stag is found in the dense forests of Dachigum National Park, Kashmir Valley and Chamba district, Himachal Pradesh.
- **Lion-Tailed Macaque, Western Ghats**
- It is the rarest, most threatened and endangered primate species found only in the Western Ghats of Southern India.
- **Purple Frog, Western Ghats**
- The purple frog, also known as the Pignose frog, is only found in the rainforests of western ghats in India. It spends most of its life underground.
- **Sangai Deer, Loktak Lake**
- It is also known as Brow Antlered Deer exclusively found in Keibul Lamjao National Park of Manipur. This park is a marshy wetland located in the southern parts of Loktak lake.
- **Nilgiri Tahr, Nilgiri Hills**
- It is a wild sheep species, endangered and endemic to the Nilgiri Hills of Western Ghats.

THREATS TO BIODIVERSITY

- Habitat loss includes habitat destruction, altering the physical environment such that a species can no longer live there, and habitat fragmentation, which involves dividing a habitat into discontinuous patches.
- [Overexploitation](#) Overexploitation involves removing organisms at a faster rate than they can be replenished. Examples include the poaching of elephants, unsustainable hunting for bush meat, overfishing, and overcollection of slow-growing plants and fungi.
- [Pollution](#) Pollution is the release of harmful chemicals or other materials into the environment. Some types of air pollution results in acid deposition and climate change. Nutrient pollution of water bodies due to fertilizer overuse results in eutrophication.
- [Invasive Species](#) Invasive species are those occurring outside of their historical distribution that cause ecological and/or economic harm. Invasive species can overpredate or outcompete native species, sometimes causing their extinction or extirpation. Examples of invasive species include the Asian carp, zebra mussels, Bd (which causes a fungal disease in amphibians), purple loosestrife, and the European starling.
- [Climate Change](#) The release of greenhouse gases, like carbon dioxide and methane, when burning fossil fuels for energy causes climate change. Not only does climate change involve an increase in average global temperature, but it also results in unpredictable weather patterns. Climate change threatens biodiversity through a variety of mechanisms and can cause species range shifts, mismatched biotic interactions, sea level rise, and ocean acidification.

Biodiversity conservation is the protection and management of biodiversity to obtain resources for sustainable development.

Biodiversity conservation has three main objectives:

To preserve the diversity of species.

Sustainable utilization of species and ecosystem.

To maintain life-supporting systems and essential ecological processes.

- **Biodiversity and its Conservation Methods**

- Biodiversity refers to the variability of life on earth. It can be conserved in the following ways:

- In-situ Conservation

- Ex-situ Conservation

- **In-situ Conservation**

- In-situ conservation of biodiversity is the conservation of species within their natural habitat. In this method, the natural ecosystem is maintained and protected.

- The in-situ conservation has several advantages. Following are the important advantages of in-situ conservation:

- It is a cost-effective and convenient method of conserving biodiversity.

- A large number of living organisms can be conserved simultaneously.

- Since the organisms are in a natural ecosystem, they can evolve better and can easily adjust to different environmental conditions.

- Certain protected areas where in-situ conservation takes place include national parks, [wildlife sanctuaries](#) and biosphere reserves.

- **National Parks**

- These are small reserves maintained by the government. Its boundaries are well demarcated and human activities such as grazing, forestry, habitat and cultivation are prohibited. For eg., Kanha National Park, and Bandipur National Park.

- **Wildlife Sanctuaries**

- These are the regions where only wild animals are found. Human activities such as timber harvesting, cultivation, collection of woods and other forest products are allowed here as long as they do not interfere with the conservation project. Also, tourists visit these places for recreation.

- **Biosphere Reserves**

- [Biosphere reserves](#) are multi-purpose protected areas where the wildlife, traditional lifestyle of the inhabitants and domesticated plants and animals are protected. Tourist and research activities are permitted here.

- **Ex-situ Conservation**
- Ex-situ conservation of biodiversity involves the breeding and maintenance of endangered species in artificial ecosystems such as zoos, nurseries, botanical gardens, gene banks, etc. There is less competition for food, water and space among the organisms.
- Ex-situ conservation has the following advantages:
- The animals are provided with a longer time and breeding activity.
- The species bred in captivity can be reintroduced in the wild.
- Genetic techniques can be used for the preservation of endangered species.

MAN-WILD LIFE CONFLICT

- Wildlife can threaten people's safety and livelihoods, which can lead to conflicts between groups of people over how to resolve the situation; experts call this 'human-wildlife conflict'.
- Human-wildlife conflicts are becoming more frequent, serious and widespread as human populations grow and habitats are lost.
- Effectively managing human-wildlife conflicts protects communities, stops conflicts escalating, builds trust in conservation and avoids retaliation against wildlife.
- Human-wildlife conflicts have unique ecological, cultural, social, historical, physical, economic and political characteristics which strategies to manage conflicts must consider.
- Human-wildlife conflict is recognised as a global concern in the UN Convention on Biological Diversity's post-2020 global biodiversity framework
- (to be adopted by Parties at CBD COP15 Part Two). Related to this, many governments are beginning to **include the management of human-wildlife conflict in national policies and strategies** to ensure resources are made available to manage them.
- There are [numerous approaches and measures that can be taken](#) to reduce the damage or impacts, de-escalate tensions, address risks to income and poverty, and develop sustainable solutions.
- These sometimes include barriers (fences, nets, trenches), guarding and early-warning systems, deterrents and repellents (sirens, lights, beehives), translocation (moving wildlife), compensation or insurance, providing risk-reducing alternatives, as well as managing tensions between stakeholders involved in these situations.
- Effective planning and implementation of such measures requires consideration of **good**
- **principles in community led-conservation**, in collaboration with the communities affected.

Values of biodiversity

- **Consumptive use value:**
- This is related to natural products that are used directly for food, fodder, timber, fuel wood etc. Humans use at least 40,000 species of plants and animals on a daily basis. Many people around the world still depend on wild species for most of their needs like food, shelter and clothing. The tribal people are completely dependent on the forests for their daily needs.
- **Ethical and Moral Value:**
- It is based on the principle of 'live and let others live'. Ethical values related to biodiversity conservation are based on the importance of protecting all forms of life. All forms of life have the right to exist on earth. Man is only a small part of the Earth's great family of species.
- Don't plants and animals have an equal right to live and exist on our planet which is like an inhabited spaceship? Morality and ethics teach us to preserve all forms of life and not to harm any organism unnecessarily.
- **Productive Use Value:**
- This is assigned to products that are commercially harvested and marketed. Almost all the present date agricultural crops have originated from wild varieties. The biotechnologists continuously use the wild species of plants for developing new, better yielding and disease resistant varieties. Biodiversity represents the original stock from which new varieties are being developed.
- **. Aesthetic Value:**
- The beauty of our planet is because of biodiversity, which otherwise would have resembled other barren planets dotted around the universe. Biological diversity adds to the quality of life and provides some of the most beautiful aspects of our existence. Biodiversity is responsible for the beauty of a landscape.
- Social value-traditional societies which had a small population and required less resources had preserved their biodiversity as a life supporting resource, modern man has rapidly depleted it even to the extent of leading to the irrecoverable loss due to extinction of several species.
- Thus apart from the local use or sale of products of biodiversity there is the social aspect in which more and more resources are used by affluent societies. The biodiversity has to a great extent been preserved by traditional societies that valued it as a resource and appreciated that its depletion would be a great loss to their society.

AQUATIC ECOSYSTEMS

- An aquatic ecosystem includes a group of interacting organisms which are dependent on one another and their water environment for nutrients and shelter. Examples of aquatic ecosystem include oceans, lakes and rivers.
- An aquatic [ecosystem](#) includes freshwater habitats like lakes, ponds, rivers, oceans and streams, wetlands, swamp, etc. and marine habitats include oceans, intertidal zone, reefs, seabed and so on. The aquatic ecosystem is the habitat for water-dependent living species including animals, plants, and microbes.

TYPES

- **Freshwater Aquatic Ecosystem**
- They cover only a small portion of earth nearly 0.8 per cent. Freshwater involves lakes, ponds, rivers and streams, wetlands, swamp, bog and temporary pools. Freshwater habitats are classified into lotic and lentic habitats. Water bodies such as lakes, ponds, pools, bogs, and other reservoirs are standing water and known as lentic habitats. Whereas lotic habitats represent flowing water bodies such as rivers, streams.
- **Lotic Ecosystems**
- They mainly refer to the rapidly flowing waters that move in a unidirectional way including the rivers and streams. These environments harbor numerous species of insects such as beetles, mayflies, stoneflies and several species of fishes including trout, eel, minnow, etc. Apart from these aquatic species, these ecosystems also include various mammals such as beavers, river dolphins and otters.
- **Lentic Ecosystems**
- They include all standing water habitats. Lakes and ponds are the main examples of Lentic Ecosystem. The word lentic mainly refers to stationary or relatively still water. These ecosystems are home to [algae](#), crabs, shrimps, amphibians such as frogs and salamanders, for both rooted and floating-leaved plants and reptiles including alligators and other water snakes are also found here.
- **Wetlands**
- Wetlands are marshy areas and are sometimes covered in water which has a wide diversity of plants and animals. Swamps, marshes, bogs

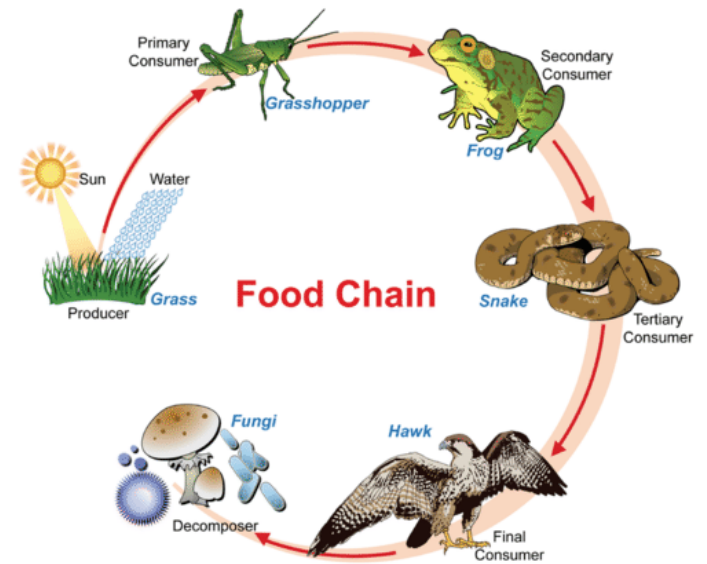
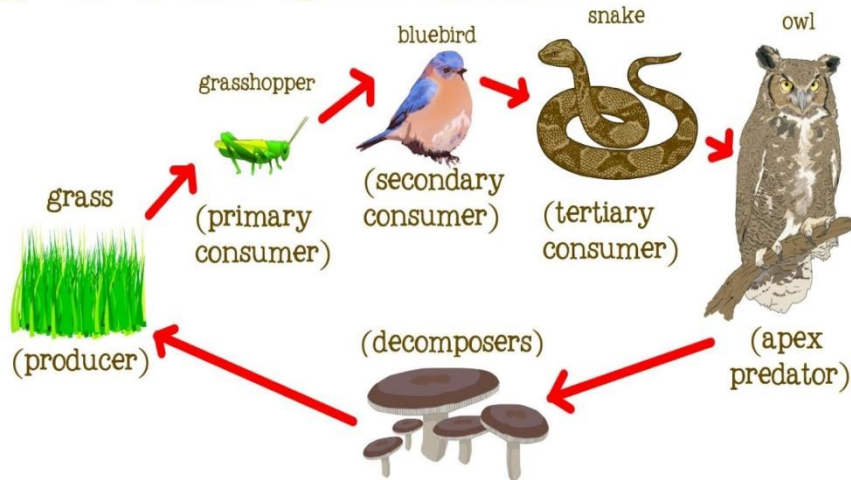
- **Marine Aquatic Ecosystem**
- Marine ecosystem covers the largest surface area of the earth. Two third of earth is covered by water and they constitute of oceans, seas, intertidal zone, reefs, seabed, estuaries, hydrothermal vents and rock pools. Each life form is unique and native to its habitat. This is because they have adaptations according to their habitat. In the case of aquatic animals, they can't survive outside of water. Exceptional cases are still there which shows another example of adaptations (e.g. mudskippers). The marine ecosystem is more concentrated with salts which make it difficult for freshwater organisms to live in. Also, marine animals cannot survive in freshwater. Their body is adapted to live in saltwater; if they are placed in less salty water, their body will swell (osmosis).
- **Ocean Ecosystems**
- Our planet earth is gifted with the five major oceans, namely Pacific, Indian, Arctic, and the Atlantic Ocean. Among all these five oceans, the Pacific and the Atlantic are the largest and deepest ocean. These oceans serve as a home to more than five lakh aquatic species. Few creatures of these ecosystems include shellfish, shark, tube worms, crab small and large ocean fishes, turtles, crustaceans, blue whale, reptiles, marine mammals, seabirds, plankton, corals and other ocean plants.

ESTUARY

- **Estuary**, partly enclosed coastal body of water in which river water is mixed with seawater. Estuaries are places where rivers meet the sea and may be defined as areas where salt water is measurably diluted with fresh water. On average, estuaries are biologically more productive than either the adjacent river or the sea, because they have a special kind of water circulation that traps plant nutrients and stimulates primary production.

FOOD CHAIN AND FOOD WEB

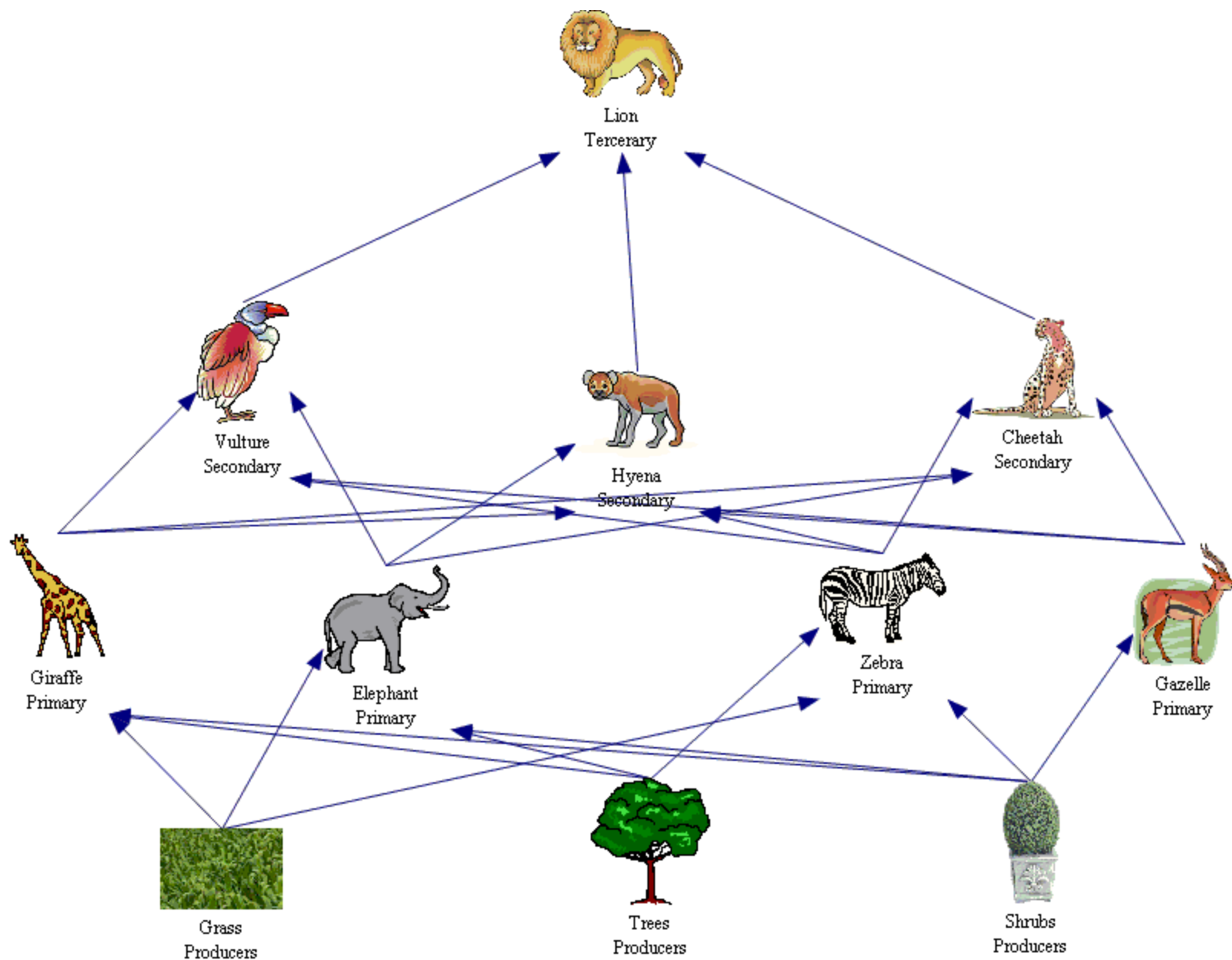
Food Chains



FOOD CHAIN AND FOOD WEB

- A food chain refers to the order of events in an ecosystem, where one living organism eats another organism, and later that organism is consumed by another larger organism. The flow of nutrients and energy from one organism to another at different trophic levels forms a food chain.
- The food chain also explains the feeding pattern or relationship between living organisms. Trophic level refers to the sequential stages in a food chain, starting with producers at the bottom, followed by primary, secondary and tertiary consumers. Every level in a food chain is known as a trophic level.

- **Food Web:**
- Several interconnected food chains form a food web. A food web is similar to a food chain but the food web is comparatively larger than a food chain. Occasionally, a single organism is consumed by many predators or it consumes several other organisms. Due to this, many trophic levels get interconnected. The food chain fails to showcase the flow of energy in the right way. But, the food web is able to show the proper representation of energy flow, as it displays the interactions between different organisms.
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- **Types of Food Chain**

- There are two types of food chains, namely the detritus food chain and the grazing food chain. Let's look at them more closely:
- **Detritus food chain:** The detritus food chain includes different species of organisms and plants like algae, bacteria, fungi, protozoa, mites, insects, worms and so on. The detritus food chain begins with dead organic material. The food energy passes into decomposers and detritivores, which are further eaten by smaller organisms like carnivores. Carnivores, like maggots, become a meal for bigger carnivores like frogs, snakes and so on. Primary consumers like fungi, bacteria, protozoans, and so on are detritivores which feed on detritus.
- **Grazing food chain:** The grazing food chain is a type of food chain that starts with green plants, passes through herbivores and then to carnivores. In a grazing food chain, energy in the lowest trophic level is acquired from photosynthesis.
- In this type of food chain, the first energy transfer is from plants to herbivores. This type of food chain depends on the flow of energy from autotrophs to herbivores. As autotrophs are the base for all ecosystems on Earth, the majority of ecosystems in the environment follow this kind of food chain.